PG No. 19 C_{3v} $3m$ (3m1 setting) [trigonal]

* Wyckoff site: `1a`, site symmetry: `3m`.

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `3m`.

No.	vector	center	mapping
1	[0, 0, Z]	[0, 0, z]	`[1,2,3,4,5,6]`

Table 2: Wyckoff bond: `3b@1a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	[X, $-X$, Z]	[0, 0, z]	`[1,4]`
2	[X, $2X$, Z]	[0, 0, z]	`[2,6]`
3	[$-2X$, $-X$, Z]	[0, 0, z]	`[3,5]`

Table 3: Wyckoff bond: `3c@1a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	[X, 0, 0]	[0, 0, z]	`[1,-5]`
2	[0, X, 0]	[0, 0, z]	`[2,-4]`
3	[$-X$, $-X$, 0]	[0, 0, z]	`[3,-6]`

Table 4: Wyckoff bond: `6d@1a`, bond symmetry: `1`

No.	vector	center	mapping
1	[X, Y, Z]	[0, 0, z]	`[1]`
2	[$-Y$, $X-Y$, Z]	[0, 0, z]	`[2]`
3	[$-X+Y$, $-X$, Z]	[0, 0, z]	`[3]`
4	[$-Y$, $-X$, Z]	[0, 0, z]	`[4]`
5	[$-X+Y$, Y, Z]	[0, 0, z]	`[5]`
6	[X, $X-Y$, Z]	[0, 0, z]	`[6]`

* Wyckoff site: `3b`, site symmetry: `.m.`

Table 5: Wyckoff bond: `3a@3b`, bond symmetry: `.m.`

No.	vector	center	mapping
1	[X, $-X$, Z]	[x, $-x$, z]	`[1,4]`
2	[X, $2X$, Z]	[x, $2x$, z]	`[2,6]`
3	[$-2X$, $-X$, Z]	[$-2x$, $-x$, z]	`[3,5]`

Table 6: Wyckoff bond: `3b@3b`, bond symmetry: `.m.`

No.	vector	center	mapping
1	[X, X, 0]	[x, $-x$, z]	`[1,-4]`

continued ...

Table 6

No.	vector	center	mapping
2	$[-X, 0, 0]$	$[x, 2x, z]$	`[2,-6]`
3	$[0, -X, 0]$	$[-2x, -x, z]$	`[3,-5]`

Table 7: Wyckoff bond: `6c@3b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, -x, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[x, 2x, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-2x, -x, z]$	`[3]`
4	$[-Y, -X, Z]$	$[x, -x, z]$	`[4]`
5	$[-X+Y, Y, Z]$	$[-2x, -x, z]$	`[5]`
6	$[X, X-Y, Z]$	$[x, 2x, z]$	`[6]`

* Wyckoff site: `6c`, site symmetry: `1`

Table 8: Wyckoff bond: `6a@6c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, z]$	`[3]`
4	$[-Y, -X, Z]$	$[-y, -x, z]$	`[4]`
5	$[-X+Y, Y, Z]$	$[-x+y, y, z]$	`[5]`
6	$[X, X-Y, Z]$	$[x, x-y, z]$	`[6]`